

Auditability, not accuracy: a head-to-head of Hybrid Confirmation Trees and Learning-to-Defer on Galaxy Zoo

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1 Framing and contribution

Human–AI decision pipelines are usually evaluated *within* a paradigm: a learning-to-defer (L2D) method against other L2D methods, or a confirmation/consultation rule against a single-human or AI-alone baseline. The **Hybrid Confirmation Tree (HCT)** of Berger et al. (HCT-1, arXiv:2602.02375; follow-up HCT-2, arXiv:2603.29866) and the L2D line of Okati, De & Gómez-Rodríguez (NeurIPS 2021, arXiv:2103.08902) and Mozannar & Sontag (ICML 2020, arXiv:2006.01862) have, to our knowledge, never been compared **head-to-head on a shared dataset, backbone, and evaluation split**. That comparison — not the Pareto plot, which is only its visualization — is the contribution here. We adopt the cost–accuracy trade-off framing of De Toni et al. (NeurIPS 2024, framing only; not implemented).

The result is a **null**: on the Galaxy Zoo binary task, with a shared strong AI backbone, *no deployable human-in-the-loop arm significantly beats AI-alone*. We report this honestly and explain why, rather than dressing up an overlapping confidence interval as a win.

2 Dataset and the non-expert-rater caveat

We use the **Galaxy Zoo binary 10k subset** (spiral vs. early-type/smooth) that Okati 2021 uses (their footnotes 8–9), sourced from the Kaggle Galaxy Zoo Challenge aggregate vote fractions.

HCT needs **real per-rater labels** (to draw two humans h_1, h_2 from a per-image urn), which Okati’s repository does not ship. We recover them by matching each Kaggle galaxy to the Willett et al. 2013 GZ2 catalog via a content-based vote-signature crosswalk (docs/HANDOFF.md §5): **5,904 of 10,000 images (59%) match** at high precision; after dropping **star_or_artifact**-mode galaxies and images with too few votes for a without-replacement draw, $N = 4,621$ survive (train 3,234 / val 693 / **test 694**; data/split_manifest.json). The matched subset is morphologically representative (binary-hardness 0.52 vs. 0.48 overall), so the 41% drop is not a difficulty-selected bias — but **59% coverage is a disclosed limitation**, not full coverage.

The raters are **Galaxy Zoo citizen-science volunteers, not domain experts** (Okati’s “human experts” is loose terminology; HANDOFF.md §2). This matters concretely: (i) the single-human *accuracy ceiling* is the volunteer crowd’s, not an astronomer’s; (ii) HCT’s tie-breaking second draw h_2 comes from the same volunteer vote distribution, so on genuinely ambiguous galaxies it is right only about as often as a coin flip. Both depress the headroom available to any human-querying arm. Crucially, the “ground truth” y is itself the **crowd-consensus majority**, so every accuracy below is *agreement with consensus, not with objective truth* (§6).

3 Methods and the invariants that make the comparison fair

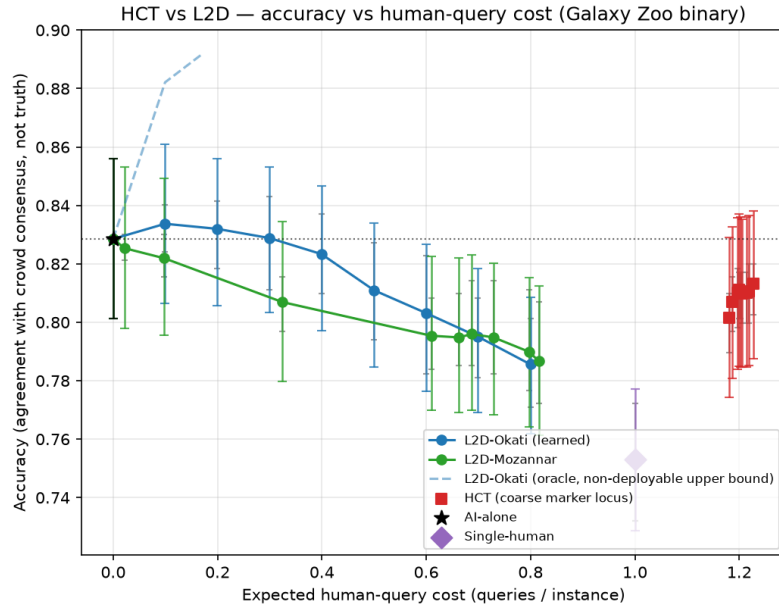
- **AI-alone** — the shared backbone thresholded at 0.5 (cost 0).
- **Single-human** — one urn draw per image (cost 1).
- **HCT** (HCT-1, Fig. 1): the human h_1 and the binarized AI decide independently; if they **agree**, accept (cost 1); if they **disagree**, draw h_2 and take its call (cost 2). Expected human cost = $1 + P(\text{human–AI disagree})$, **bounded in $[1, 2]$** , swept via the AI binarization threshold θ — the *only* HCT knob. HCT is a decision rule, not a trained model.
- **L2D-Okati** (Thm. 3): for a fixed classifier, the optimal triage thresholds the per-instance gap between expected machine and human loss; a budget b caps the deferral fraction. Cost = deferral fraction $\in [0, 1]$.

- **L2D-Mozannar** (eq. 10): the consistent convex surrogate L_{CE}^α ; the deferral knob α sweeps a smooth curve. Cost = deferral fraction $\in [0, 1]$.

Shared-backbone + frozen-split invariants (GROUND_TRUTH.md §7; HANDOFF.md §10): every arm consumes the *same* committed backbone scores/embeddings and the *same* seed=0 70/15/15 split (hash guarded by `make verify-split`); one shared metrics module computes all numbers.

Deviation — Option B (logged, docs/DECISIONS.md). The L2D papers *co-train* classifier and rejector. We instead **freeze the shared backbone and learn only the rejector/triage** on top of its embeddings. This is deliberate: co-training would give each L2D arm a *different* classifier, confounding the comparison with backbone differences. Freezing makes the AI identical across all arms, so any measured gap is attributable to the deferral policy alone — it **removes a confound rather than adding one**. The cost is that we measure these L2D methods’ *triage* quality, not their end-to-end co-training; we do not import their co-trained accuracy claims here.

4 Results



HCT is a coarse marker locus (cost in [1,2], swept via the AI threshold); the L2D arms are smooth lines (cost in [0,1], swept via their deferral parameter) — a genuine shape asymmetry, not an artefact. The Okati oracle (dashed) uses test labels and is a non-deployable upper bound. Accuracy is agreement with the crowd-consensus label, not objective truth. Error bars = bootstrap CI over the 694 test instances; whiskers = min/max across rater-draw seeds. Regions where an arm’s CI overlaps AI-alone are not significant differences.

Figure 1: HCT vs L2D — accuracy vs expected human-query cost on the Galaxy Zoo binary subset.

All numbers from `results/tables/summary.csv`; CIs are 95% bootstrap over the 694 test instances (stochastic arms also resampled across rater-draw seeds [0–4]). The frontier is shown in Figure 1.

Arm (policy)	Accuracy [95% CI]	Cost	vs AI-alone	Deployable
AI-alone	0.829 [0.801, 0.856]	0	— (baseline)	yes
L2D-Okati (learned), $b=0.1$	0.834 [0.806, 0.861]	0.10	ns (CI overlaps)	yes
L2D-Mozannar (best deferral, $\alpha=0.95$)	0.822 [0.796, 0.849]	0.10	ns (CI overlaps), below mean	yes
HCT ($\theta=0.1$)	0.813 [0.788, 0.838]	1.23	ns (CI overlaps), below mean	yes
Single-human	0.753 [0.729, 0.777]	1.00	below (CI excludes 0.829)	yes
L2D-Okati (oracle), $b\approx 0.2$	0.892 [0.871, 0.911]	0.17	above — but non-deployable	no

Curve-shape and cost-regime asymmetry (stated explicitly, never obscured). HCT is a **coarse marker locus**, drawn as discrete markers — sweeping θ moves its expected cost only within a narrow band (observed 1.18–1.23 here) because cost is $1 + P(\text{disagree})$ and is structurally **bounded in** $[1, 2]$. The L2D arms are **smooth lines** whose cost ranges over $[0, 1]$. The two methods therefore live in **disjoint cost regimes** (L2D in $[0, 1]$, HCT in $[1, 2]$) and are *not* mutually Pareto-comparable point-for-point; HCT is never drawn as a continuous curve. This is a genuine property of the methods, not an artefact of plotting.

Reading the arms. L2D-Okati(learned) peaks at 0.834 @ cost 0.10 then declines as more (often correct) machine decisions are handed to noisier humans; Mozannar’s best deferral point (0.822) sits *below* AI-alone in the mean (its CI still overlaps) and degrades further with deferral. HCT’s accuracy is essentially flat (~ 0.81) across θ and sits below AI-alone in the mean. The Okati **oracle** (dashed) — which peeks at test labels to triage perfectly — climbs to 0.892, but it is a **non-deployable upper bound** and is excluded from the deployable frontier.

5 Central finding (an honest null)

No deployable arm significantly beats AI-alone. The verdict is computed mechanically in eval (any_deployable_arm_beats_ai_alone = false, figure manifest): every deployable arm’s CI either **overlaps** AI-alone (L2D-Okati-learned, HCT, Mozannar) or sits **strictly below** it (single-human is the only arm whose CI excludes 0.829 — it is significantly *worse*). L2D-Okati-learned’s +0.5 pt mean at cost 0.10 is within noise; HCT’s mean is below AI-alone at cost > 1 .

Interpretation: **on consensus-labeled Galaxy Zoo with a strong AI, human deferral buys auditability, not accuracy.** HCT’s value is that *at least one human always approves the deployed decision* at a bounded, transparent, training-free cost — a governance property, not an accuracy gain. The oracle shows ~ 6 pts of headroom *exists* in principle, but it is **unrealizable** with a deployable rejector on these frozen embeddings: the learned triage cannot identify which machine errors a volunteer would fix.

Upstream self-claims are not findings here. HCT-1 reports 28–44% cost savings *on its own datasets*; that is a property of those datasets, **not** a result on Galaxy Zoo, and we do not import it. Likewise we measure the L2D arms’ triage over a frozen backbone (Option B), not their co-trained end-to-end numbers. Every claim in this report traces to an artifact under **results/**.

6 Limitations

- **Single dataset, single backbone.** One task (Galaxy Zoo binary), one frozen ResNet-50. No claim generalizes beyond this setting.
- **Accuracy = agreement with crowd consensus, not truth** (HANDOFF.md §9). Because y is the volunteer majority, a single human’s accuracy is the majority vote share by construction, and HCT’s h_2 is $\sim 50\%$ on near-split galaxies by construction — this compresses inter-arm gaps and is a *threat to the comparison itself*, not a footnote.
- **Non-expert raters** (§2): the human ceiling and the h_2 tiebreak are volunteer-quality.
- **59% crosswalk coverage** (§2): 4,621 of 10,000; disclosed, representative, but not complete.
- **Urn structural limits** (HANDOFF.md §6): $h_1 \perp h_2 \mid x$ is enforced (both are independent draws from the same per-image distribution), so h_2 is a **second draw, not a second human**; the positive human–human correlation Berger studies lives only in between-image urn-composition variance, and the opinion-leader (ρ) reconstruction is **not supported** by aggregate counts.
- **Option-B deviation** (§3): triage-only, not the papers’ co-training; removes a backbone confound but does not evaluate end-to-end co-training.
- **Reproducibility boundary** (docs/REPRODUCE.md): make all regenerates this report’s figure and table deterministically on CPU from the committed frozen artifacts; the backbone training + embedding export (GPU/Colab, notebooks/m2_colab.ipynb) and the raw-catalog label build are upstream of that interface and are not re-run by make all.